

Verified Statements for a Finite-Precision Lanczos Analysis

A Formal Statement Record

Anonymous formalization report

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Abstract

This document gives a mathematical rendering of the principal statements checked in a Lean 4 development of finite-precision Lanczos analysis. The formal input is not an implementation of floating-point arithmetic. It is a real symmetric matrix together with computed vectors and coefficients that satisfy explicitly stated local arithmetic identities and error bounds. From that interface the development verifies explicit local-error estimates, a Paige-type commutator identity and loss-of-orthogonality bound, Ritz-value containment and stabilization, and a Greenbaum-type backward model. The backward model is a symmetric tridiagonal matrix whose leading block is the computed Lanczos tridiagonal, which reproduces that block under exact Lanczos, and whose spectrum and starting spectral measure are controlled relative to the original matrix. Every constant and hypothesis used below is displayed. The purpose is auditability: the theorem names attached to the statements identify the corresponding Lean declarations.

1 Purpose, scope, and formal status

The authoritative objects are the Lean sources in the accompanying project. This manuscript translates their paper-facing declarations into conventional numerical-linear-algebra notation. It is deliberately conservative: no asymptotic notation is substituted for a proved constant, and conclusions that are conditional in Lean remain conditional here.

The project builds with Lean 4 and Mathlib using the pinned toolchain v4.32.2. A full `lake build` completed successfully. An axiom audit of the principal results reported only `propext`, `Classical.choice`, and `Quot.sound`, the standard logical axioms used by Lean and Mathlib. A source search found no occurrences of `sorry`, `admit`, or user-declared axioms.

Central modelling limitation. The Lean development does not prove that executing a particular machine-level Algorithm 1 produces the formal run described below. It proves that *any* family of real objects satisfying Assumption 1 obeys the conclusions in this document. Connecting an implementation, a floating-point semantics, or a probabilistic rounding model to that interface is a separate obligation.

2 Notation

All matrices and vectors are real. The vector norm and matrix norm are the Euclidean norm and its induced operator 2-norm, respectively. Transpose is written T , and e_j denotes a coordinate vector. The indices in the mathematical presentation are one-based. The formal development uses natural indices with the phantom values $v_0 = 0$ and $\beta_0 = 0$.

For a real symmetric matrix $H \in \mathbb{R}^{n \times n}$, write

$$\lambda_{\min}(H) = \min \text{spec}(H), \quad \lambda_{\max}(H) = \max \text{spec}(H), \quad d_H(\theta) = \text{dist}(\theta, \text{spec}(H)).$$

The formal definitions return zero outside the symmetric, positive-dimensional case; Assumption 1 implies symmetry, and the run assumptions imply $n > 0$, so only the usual spectral interpretation occurs in the results below.

For $x \in \mathbb{R}^n$ and a symmetric A , define its starting spectral mass by

$$\mu_{A,x} = \sum_i (u_i^T x)^2 \delta_{\lambda_i(A)}, \quad (1)$$

where (u_i) is the orthonormal eigenbasis selected by the formal spectral theorem. Repeated eigenvalues are combined because the formal measure is a finitely supported function on $\mathbb{R}$.

For finite atomic masses μ and ν , the development defines

$$\mathcal{W}_1^{\text{dual}}(\mu, \nu) = \sup_{\text{Lip}(f) \leq 1} \left| \int f d\mu - \int f d\nu \right|. \quad (2)$$

This is the dual quantity named `atomicWasserstein` in the code. We retain the superscript “dual” here to make the formal definition explicit.

3 The formal finite-precision interface

Assumption 1 (Lanczos run interface). *Let $n, k \in \mathbb{N}$. The input to every run-level result consists of*

$$H \in \mathbb{R}^{n \times n}, \quad v_j, u_j, w_j, z_j, e_j^u, e_j^w, e_j^z, \Delta_j \in \mathbb{R}^n, \quad \alpha_j, \beta_j, e_j^\alpha \in \mathbb{R}, \quad \varepsilon \in \mathbb{R},$$

subject to all of the following conditions.

(A1) $H^T = H$, $\varepsilon \geq 0$, $k > 0$, and $k\varepsilon \leq 1/100$.

(A2) The phantom entries satisfy $v_0 = 0$ and $\beta_0 = 0$.

(A3) For every $j = 0, \dots, k-1$,

$$u_{j+1} = H v_{j+1} + e_{j+1}^u, \quad \|e_{j+1}^u\|_2 \leq \varepsilon \|H\|_2 \|v_{j+1}\|_2, \quad (3)$$

$$w_{j+1} = u_{j+1} - \beta_j v_j + e_{j+1}^w, \quad \|e_{j+1}^w\|_2 \leq \varepsilon (\|u_{j+1}\|_2 + |\beta_j| \|v_j\|_2), \quad (4)$$

$$\alpha_{j+1} = v_{j+1}^T w_{j+1} + e_{j+1}^\alpha, \quad |e_{j+1}^\alpha| \leq \varepsilon \|v_{j+1}\|_2 \|w_{j+1}\|_2, \quad (5)$$

$$z_{j+1} = w_{j+1} - \alpha_{j+1} v_{j+1} + e_{j+1}^z, \quad \|e_{j+1}^z\|_2 \leq \varepsilon (\|w_{j+1}\|_2 + |\alpha_{j+1}| \|v_{j+1}\|_2), \quad (6)$$

$$\beta_{j+1} v_{j+2} = z_{j+1} + \Delta_{j+1}, \quad \|\Delta_{j+1}\|_2 \leq \varepsilon \|z_{j+1}\|_2. \quad (7)$$

(A4) The computed Lanczos vectors are nearly normalized:

$$|\|v_j\|_2^2 - 1| \leq \varepsilon, \quad 1 \leq j \leq k.$$

If $\beta_k \neq 0$, the additional vector satisfies $|\|v_{k+1}\|_2^2 - 1| \leq \varepsilon$.

(A5) $\beta_j \geq 0$ for $0 \leq j \leq k$, and $\beta_j > 0$ for $1 \leq j < k$.

Formal declaration(s):

`FinitePrecisionLanczos.LanczosRun`.

There are no other numerical assumptions at the run interface. In particular, $k \leq n$ is not assumed, β_k need not be positive, and no orthogonality between distinct v_i and v_j is

assumed. The condition $\beta_j > 0$ only excludes an internal exact breakdown before the last reported step.

Define, for $1 \leq j \leq k$,

$$f_j = Hv_j - \beta_{j-1}v_{j-1} - \alpha_jv_j - \beta_jv_{j+1}, \quad (8)$$

$$g_j = v_j^T v_j - 1, \quad p_j = v_j^T (\beta_j v_{j+1}), \quad p_0 = 0. \quad (9)$$

Let

$$V = [v_1 \ \cdots \ v_k], \quad F = [f_1 \ \cdots \ f_k],$$

and let T_k be the symmetric tridiagonal matrix with diagonal $(\alpha_1, \dots, \alpha_k)$ and sub/super-diagonal $(\beta_1, \dots, \beta_{k-1})$. Finally set

$$b = \beta_k v_{k+1}, \quad B = be_k^T.$$

4 Local errors and assembled recurrence

Theorem 4.1 (Explicit local errors). *Under Assumption 1, for every $1 \leq j \leq k$,*

$$\|f_j\|_2 \leq 700j\varepsilon\|H\|_2, \quad |g_j| \leq \varepsilon, \quad |p_j| \leq 1200j\varepsilon\|H\|_2. \quad (10)$$

Moreover,

$$HV = VT_k + B + F, \quad \|F\|_2 \leq 700k^2\varepsilon\|H\|_2. \quad (11)$$

Formal declaration(s):

`LanczosRun.local_errors`; `LanczosRun.governing_matrix`; `LanczosRun.Fmat_bound`.

For audit purposes, the local proof also verifies the componentwise estimates

$$\|e_j^u\|_2 \leq 2j\varepsilon\|H\|_2, \quad \|e_j^w\|_2 \leq 200j\varepsilon\|H\|_2, \quad |e_j^\alpha| \leq 200j\varepsilon\|H\|_2, \quad (12)$$

and

$$\|e_j^z\|_2 \leq 300j\varepsilon\|H\|_2, \quad \|\Delta_j\|_2 \leq 100j\varepsilon\|H\|_2. \quad (13)$$

These are the declarations `LanczosRun.eu_explicit`, `LanczosRun.ew_explicit`, `LanczosRun.ealpha_explicit`, `LanczosRun.ez_explicit`, and `LanczosRun.delta_explicit`.

5 Gram commutator and Paige bounds

Set

$$\Omega = V^T V - I, \quad D = \text{diag}(g_1, \dots, g_k),$$

and let R be the strictly upper-triangular part of Ω . Thus

$$\Omega = R^T + D + R. \quad (14)$$

Introduce

$$\begin{aligned} C_b &= V^T B, & G &= V^T F - F^T V, \\ Q &= T_k D - D T_k, & N &= \text{diag}(p_0 - p_1, p_1 - p_2, \dots, p_{k-1} - p_k). \end{aligned}$$

For a matrix M , write M_{up} for its strictly upper-triangular part.

Theorem 5.1 (Gram commutator). *Under Assumption 1,*

$$T_k \Omega - \Omega T_k = C_b - C_b^T + G, \quad (15)$$

and every entry of G satisfies

$$|G_{rs}| \leq 2800k \varepsilon \|H\|_2. \quad (16)$$

The upper-triangular commutator has the exact form

$$T_k R - R T_k = C_b + N + G_{\text{up}} - Q_{\text{up}}. \quad (17)$$

Furthermore, Q_{up} is supported on the first superdiagonal and

$$|(Q_{\text{up}})_{rs}| \leq 600k \varepsilon \|H\|_2. \quad (18)$$

Formal declaration(s):

`LanczosRun.gram_commutator_conclusion.`

For a unit vector $y \in \mathbb{R}^k$ satisfying $T_k y = \theta y$, let $x = Vy$ be its computed Ritz vector. The last coordinate of y is denoted y_k .

Theorem 5.2 (Paige boundary-product bound). *For every unit eigenpair (θ, y) of T_k ,*

$$|(x^T b)y_k| \leq 6000k^3 \varepsilon \|H\|_2. \quad (19)$$

Formal declaration(s):

`LanczosRun.paige_bound.`

The theorem is an absolute product bound. It does not separately bound $\|b\|_2$, $|y_k|$, or the angle between x and b .

6 Ritz-value containment and stabilization

Define the three nonnegative scales

$$E_k = 6000k^3 \varepsilon \|H\|_2, \quad (20)$$

$$F_k = 700k^2 \varepsilon \|H\|_2, \quad (21)$$

$$\Psi_k = 8kE_k = 48000k^4 \varepsilon \|H\|_2, \quad (22)$$

and, for $1 \leq t \leq k$,

$$\mathcal{R}(t) = (8t^2 + 5)E_k + 3F_k. \quad (23)$$

The prefix T_t below is the leading $t \times t$ principal submatrix of the computed tridiagonal T_k .

Theorem 6.1 (Containment for every prefix). *For every $1 \leq t \leq k$ and every unit eigenpair $T_t y = \theta y$,*

$$\lambda_{\min}(H) - \mathcal{R}(t) \leq \theta \leq \lambda_{\max}(H) + \mathcal{R}(t). \quad (24)$$

In particular, this holds with $t = k$ for every eigenvalue of T_k .

Formal declaration(s):

`LanczosRun.containment_prefix; LanczosRun.containment; LanczosRun.containment_eigenvalue.`

The proof in Lean does not assume a small Gram defect. It uses a formally verified descent argument when $|y^T R y| > 3/8$. In the complementary case, the following sharper local statement is available.

Proposition 6.2 (Small-loss localization). *If $T_k y = \theta y$, $y^T y = 1$, and $|y^T R y| \leq 3/8$, then*

$$d_H(\theta) \leq 3\beta_k |y_k| + 3F_k. \quad (25)$$

Formal declaration(s):

`LanczosRun.small_loss_localization.`

Theorem 6.3 (Stabilized Ritz localization). *For every unit eigenpair $T_k y = \theta y$,*

$$d_H(\theta) \leq (k-1)\Psi_k + 3\max\{\beta_k |y_k|, \Psi_k\} + 3F_k. \quad (26)$$

Consequently,

$$d_H(\theta) \leq 3\beta_k |y_k| + (k+2)\Psi_k + 3F_k. \quad (27)$$

If either $\beta_k |y_k| \leq \Psi_k$ or $\beta_k = 0$, then

$$d_H(\theta) \leq (k+2)\Psi_k + 3F_k. \quad (28)$$

The terminal statement also holds directly for every eigenvalue of T_k when $\beta_k = 0$.

Formal declaration(s):

`LanczosRun.stabilized_max_form; LanczosRun.stabilized; LanczosRun.stabilized_of_boundary_le; LanczosRun.terminal_localization_eigenvalue.`

Because $k > 0$, the displayed coefficient $k+2$ is exactly the mathematical simplification of the formal natural-number expression $(k-1) + 3$.

7 Normalization and the Greenbaum construction

Let $d_j = \|v_j\|_2$ for $1 \leq j \leq k$, $D_v = \text{diag}(d_1, \dots, d_k)$, and

$$\bar{V} = V D_v^{-1} = [\bar{v}_1 \ \cdots \ \bar{v}_k].$$

The run assumptions imply $d_j > 0$, so this normalization is well-defined, and $\|\bar{v}_j\|_2 = 1$. If $\beta_k \neq 0$, set

$$d_{k+1} = \|v_{k+1}\|_2, \quad \bar{v}_{k+1} = v_{k+1}/d_{k+1}, \quad \bar{\beta}_k = \beta_k d_{k+1}/d_k.$$

If $\beta_k = 0$, the formal definitions use $\bar{v}_{k+1} = \bar{v}_1$ and $\bar{\beta}_k = 0$.

Theorem 7.1 (Normalized local recurrence). *There is a matrix $\bar{E} \in \mathbb{R}^{n \times k}$ such that*

$$H\bar{V} = \bar{V}T_k + \bar{\beta}_k \bar{v}_{k+1} e_k^T + \bar{E}, \quad (29)$$

with

$$\|\bar{E}\|_2 \leq 2600k^4 \varepsilon \|H\|_2. \quad (30)$$

For $1 \leq j < k$,

$$\beta_j |\bar{v}_j^T \bar{v}_{j+1}| \leq 4800k\varepsilon \|H\|_2, \quad (31)$$

and at the boundary

$$\bar{\beta}_k |\bar{v}_k^T \bar{v}_{k+1}| \leq 4800k\varepsilon \|H\|_2. \quad (32)$$

Formal declaration(s):

`LanczosRun.normalized_governing; LanczosRun.normalized_local_errors.`

Let U be the strictly upper-triangular part of $\bar{V}^T \bar{V}$. Since the columns of $\bar{V}$ are unit vectors,

$$\bar{V}^T \bar{V} = I + U + U^T. \quad (33)$$

Set $a = \bar{V}^T \bar{v}_{k+1}$ and

$$J = T_k U - U T_k - \bar{\beta}_k e_k a^T. \quad (34)$$

Theorem 7.2 (Normalized commutator). *The matrix J is upper triangular,*

$$J - J^T = \bar{V}^T \bar{E} - \bar{E}^T \bar{V}, \quad (35)$$

and

$$\|J\|_2 \leq 50000k^5\varepsilon\|H\|_2. \quad (36)$$

Formal declaration(s):

`LanczosRun.normalized_gram_commutator.`

Define the finite geometric transform

$$C = \sum_{r=0}^{k-1} (-U)^r, \quad K = I - C, \quad (37)$$

and set

$$s = Ca, qquad y_0 = \bar{v}_{k+1} - \bar{V}Cs. \quad (38)$$

Theorem 7.3 (Nilpotent isometry identities). *The matrix K is strictly upper triangular and*

$$K^k = 0, \quad \|K\|_2 \leq 1, \quad \|C\|_2 \leq 2. \quad (39)$$

Moreover,

$$(\bar{V}C)^T(\bar{V}C) = I - K^T K, \quad (40)$$

$$(\bar{V}C)^T y_0 = -K^T s, \quad (41)$$

$$\|s\|_2^2 + \|y_0\|_2^2 = 1. \quad (42)$$

Formal declaration(s):

`LanczosRun.nilpotent_isometry.`

The notation $\|s\|_2^2$ in (42) means $(\|s\|_2)^2$; both s and y_0 use Euclidean norms.

8 Physical dilation and exact backward model

Let

$$\mathcal{H} = \text{diag}(H, \dots, H) \in \mathbb{R}^{nk \times nk} \quad (43)$$

contain k copies of H . Form $X \in \mathbb{R}^{nk \times k}$ by stacking the k blocks

$$\bar{V}C, \bar{V}CK, \dots, \bar{V}CK^{k-1}. \quad (44)$$

The boundary vector $q \in \mathbb{R}^{nk}$ has first block y_0 and, for $r = 1, \dots, k-1$, block $r+1$ equal to $\bar{V}CK^{r-1}s$.

Theorem 8.1 (Physical dilation). *The scalar $\bar{\beta}_k$ is nonnegative and*

$$X^T X = I, qquad X^T q = 0, qquad \|q\|_2 \leq 1. \quad (45)$$

The first column of X is $(\bar{v}_1^T, 0, \dots, 0)^T$. There is $\hat{E} \in \mathbb{R}^{nk \times k}$ such that

$$\mathcal{H}X = XT_k + \bar{\beta}_k q e_k^T + \hat{E}, \quad (46)$$

and

$$\|\hat{E}\|_2 \leq 10^6 k^8 \varepsilon \|H\|_2. \quad (47)$$

At the block level, each of the k propagated recurrence errors is bounded by $10^6 k^7 \varepsilon \|H\|_2$.

Formal declaration(s):

`LanczosRun.physical_dilation; LanczosRun.block_recurrence.`

The formal correction is a symmetric matrix $\hat{\Delta} \in \mathbb{R}^{nk \times nk}$ satisfying

$$\hat{\Delta}^T = \hat{\Delta}, \text{quad}\hat{\Delta}X = -\hat{E}, \text{quad}\|\hat{\Delta}\|_2 \leq 3\|\hat{E}\|_2 \leq 3,000,000k^8\varepsilon\|H\|_2. \quad (48)$$

Consequently, $\hat{H} = \mathcal{H} + \hat{\Delta}$ is symmetric and obeys the exact recurrence

$$\hat{H}X = XT_k + \bar{\beta}_k q e_k^T. \quad (49)$$

Formal declaration(s):

`LanczosRun.greenbaumDelta_transpose; LanczosRun.greenbaumDelta_mul_X; LanczosRun.greenbaumDelta_bound;`
`LanczosRun.greenbaum_corrected_recurrence.`

Theorem 8.2 (Existence of the Greenbaum model). *There exist an integer $d \geq 0$, an orthogonal square matrix*

$$Q \in \mathbb{R}^{nk \times (k+d)}, \quad nk = k + d, \text{quad}Q^T Q = I, \text{quad}Q Q^T = I,$$

and a real symmetric tridiagonal matrix $\tilde{T} \in \mathbb{R}^{(k+d) \times (k+d)}$ with the following properties.

(G1) *The first k columns of Q are the columns of X .*

(G2) *The leading $k \times k$ block of $\tilde{T}$ is exactly T_k .*

(G3) *With*

$$S = Q^T \mathcal{H} Q, \quad (50)$$

one has

$$\delta := \|S - \tilde{T}\|_2 \leq 3\|\hat{E}\|_2 \leq 3,000,000k^8\varepsilon\|H\|_2. \quad (51)$$

(G4) *The first $k - 1$ prescribed subdiagonal entries of $\tilde{T}$ are $\beta_1, \dots, \beta_{k-1}$ and are positive.*

Formal declaration(s):

`LanczosRun.exists_greenbaumModel; LanczosRun.GreenbaumModel.delta_explicit; LanczosRun.GreenbaumModel.`
`prescribed_subdiagonal_positive.`

Here “tridiagonal” follows from the conjunction of formal symmetry and the formal upper-tridiagonal predicate. The theorem does not assert positivity of the subdiagonal beyond the leading $k \times k$ block.

Theorem 8.3 (Exact Lanczos reproduction). *Let exact Lanczos be initialized at $e_1 \in \mathbb{R}^{k+d}$ and applied to $\tilde{T}$ for k steps. In the precise sense of the exact-run interface in Definition 8.4, an exact run exists, and every such run returns the tridiagonal matrix T_k . In particular, its computed $\alpha_1, \dots, \alpha_k$ and internal $\beta_1, \dots, \beta_{k-1}$ agree with those of the original formal run.*

Formal declaration(s):

`LanczosRun.GreenbaumModel.exactLanczosRun; LanczosRun.GreenbaumModel.exact_lanczos_alpha; LanczosRun.`
`GreenbaumModel.exact_lanczos_beta; LanczosRun.GreenbaumModel.exact_lanczos_returns_Tmat.`

Definition 8.4 (Exact-run interface). For a symmetric matrix A , starting vector q , and number of steps k , the formal exact run contains sequences v_j, α_j, β_j satisfying

$$v_0 = 0, \quad \beta_0 = 0, \quad v_1 = q, \quad \|v_{j+1}\|_2 = 1 \quad (0 \leq j < k),$$

$$\alpha_{j+1} = v_{j+1}^T A v_{j+1},$$

and, for $j + 1 < k$,

$$\beta_{j+1} > 0, \text{quad} \beta_{j+1} v_{j+2} = A v_{j+1} - \alpha_{j+1} v_{j+1} - \beta_j v_j.$$

This is the exact scope of Theorem 8.3; no particular software implementation of exact Lanczos is invoked.

Formal declaration(s):

`FinitePrecisionLanczos.ExactLanczos.`

9 Spectral and starting-measure consequences

Let $q_H = \bar{v}_1$ and $q_M = e_1 \in \mathbb{R}^{k+d}$. The matrix S in (50) is orthogonally similar to k copies of H .

Theorem 9.1 (Exact starting moments for the unperturbed dilation). *For every $m \in \mathbb{N}$,*

$$q_M^T S^m q_M = q_H^T H^m q_H. \quad (52)$$

Equivalently, the finite atomic starting spectral masses agree exactly:

$$\mu_{S, q_M} = \mu_{H, q_H}. \quad (53)$$

Formal declaration(s):

`LanczosRun.GreenbaumModel.starting_moment; LanczosRun.GreenbaumModel.starting_spectralMass.`

The exact equality is between H and S , not between H and $\tilde{T}$. The latter comparison is perturbative, as in the next two theorems.

Theorem 9.2 (Localization of every model eigenvalue). *For every $\tilde{\lambda} \in \text{spec}(\tilde{T})$,*

$$d_H(\tilde{\lambda}) \leq \delta \leq 3,000,000 k^8 \varepsilon \|H\|_2. \quad (54)$$

Formal declaration(s):

`LanczosRun.GreenbaumModel.eigenvalue_localization; LanczosRun.GreenbaumModel.eigenvalue_localization_explicit.`

Theorem 9.3 (Starting spectral mass in dual Wasserstein distance). *The physical and model starting masses satisfy*

$$\mathcal{W}_1^{\text{dual}}(\mu_{H, q_H}, \mu_{\tilde{T}, q_M}) \leq \sqrt{nk} \delta \leq \sqrt{nk} 3,000,000 k^8 \varepsilon \|H\|_2. \quad (55)$$

Formal declaration(s):

`LanczosRun.GreenbaumModel.starting_wasserstein; LanczosRun.GreenbaumModel.starting_wasserstein_explicit.`

We finally state the proved cluster-mass estimate in full. Let λ_i and $w_i = (u_i^T q_H)^2$ be the physical eigenvalues and starting weights. Let $\tilde{\lambda}_j$ and $\tilde{w}_j = (\tilde{u}_j^T q_M)^2$ be their model counterparts. For a predicate G on $\mathbb{R}$, define its δ -neighborhood relative to the physical eigenvalue list by

$$C_G = \{t \in \mathbb{R} : \text{there exists } i \text{ with } G(\lambda_i) \text{ and } |t - \lambda_i| \leq \delta\}. \quad (56)$$

Theorem 9.4 (Starting cluster mass). *Assume that G and membership in C_G are decidable. Suppose there is $g \in \mathbb{R}$ such that every pair of physical eigenvalues lying on opposite sides of G satisfies*

$$g \leq |\lambda_i - \lambda_j|, \quad (57)$$

and assume $2\delta < g$. Then

$$(g - 2\delta) \left| \sum_{i:G(\lambda_i)} w_i - \sum_{j:\tilde{\lambda}_j \in C_G} \tilde{w}_j \right| \leq \sqrt{nk} \delta, \quad (58)$$

and hence

$$\left| \sum_{i:G(\lambda_i)} w_i - \sum_{j:\tilde{\lambda}_j \in C_G} \tilde{w}_j \right| \leq \frac{\sqrt{nk} \delta}{g - 2\delta}. \quad (59)$$

Formal declaration(s):

`LanczosRun.GreenbaumModel.starting_cluster_mass_product;` `LanczosRun.GreenbaumModel.starting_cluster_mass.`

The theorem uses the actual model error δ , not merely its explicit upper bound. Substituting the bound in (51) into a denominator would require separately checking that the resulting denominator remains positive; that substitution is not packaged as a Lean theorem.

10 What is, and is not, established

The verified implication can be summarized as

$$\boxed{\text{formal run interface}} \implies \begin{array}{l} \text{local bounds and exact matrix recurrence} \\ \text{Paige commutator, loss, containment, stabilization} \\ \text{normalized recurrence and physical dilation} \\ \text{symmetric tridiagonal backward model} \\ \text{exact reproduction and spectral-measure bounds.} \end{array}$$

The following claims are intentionally *not* made by the formalization.

1. It does not derive Assumption 1 from source code, hardware floating-point operations, IEEE 754, or a compiler semantics.
2. It does not identify ε with a particular unit roundoff; the five local error bounds in Assumption 1 are hypotheses.
3. It does not assume or prove $k \leq n$.
4. It does not claim exact equality of the starting spectral masses of H and $\tilde{T}$; exact equality holds for H and the unperturbed dilation S , while (55) controls $\tilde{T}$.
5. It does not claim that all subdiagonal entries of the completed $\tilde{T}$ are positive. Positivity is verified only for the first $k - 1$ entries inherited from the computed run.
6. The constants are explicit sufficient bounds proved in Lean. No claim of sharpness or optimal dependence on k is made.

11 Declaration correspondence

Table 1 is a compact audit map from the displayed mathematical results to the principal Lean declarations. Supporting algebraic, finite-dimensional spectral, tridiagonal-completion, and finite-atomic-measure lemmas are imported by these declarations but are not individually restated as journal theorems here.

Mathematical item	Lean declaration(s)
Run assumptions	<code>FinitePrecisionLanczos.LanczosRun</code>
Local errors	<code>LanczosRun.local_errors</code>
Assembled recurrence	<code>LanczosRun.governing_matrix</code> , <code>LanczosRun.Fmat_bound</code>
Gram commutator	<code>LanczosRun.gram_commutator_conclusion</code>
Paige bound	<code>LanczosRun.paige_bound</code>
Prefix containment	<code>LanczosRun.containment_prefix</code> , <code>LanczosRun.containment</code>
Stabilization	<code>LanczosRun.stabilized_max_form</code> , <code>LanczosRun.stabilized</code>
Normalized errors	<code>LanczosRun.normalized_local_errors</code>
Normalized commutator	<code>LanczosRun.normalized_gram_commutator</code>
Nilpotent transform	<code>LanczosRun.nilpotent_isometry</code>
Physical dilation	<code>LanczosRun.physical_dilation</code>
Symmetric correction	<code>LanczosRun.greenbaumDelta_bound</code> , <code>LanczosRun.greenbaum_corrected_recurrence</code>
Backward model existence	<code>LanczosRun.exists_greenbaumModel</code>
Model error	<code>LanczosRun.GreenbaumModel.delta_explicit</code>
Exact reproduction	<code>LanczosRun.GreenbaumModel.exact_lanczos_returns_Tmat</code>
Starting moments/mass	<code>LanczosRun.GreenbaumModel.starting_moment</code> , <code>LanczosRun.GreenbaumModel.starting_spectralMass</code>
Eigenvalue localization	<code>LanczosRun.GreenbaumModel.eigenvalue_localization_explicit</code>
Dual Wasserstein bound	<code>LanczosRun.GreenbaumModel.starting_wasserstein_explicit</code>
Cluster mass	<code>LanczosRun.GreenbaumModel.starting_cluster_mass</code>

Table 1: Principal correspondence between this statement record and Lean.

12 Conclusion

Subject exactly to Assumption 1, the formal development verifies a complete chain from local floating-point-style identities to explicit Paige and Greenbaum conclusions. The central backward object is a symmetric tridiagonal $\tilde{T}$ with computed leading block T_k and operator-norm distance at most $3,000,000k^8\varepsilon\|H\|_2$ from an orthogonal copy of k repeated blocks of H . Exact Lanczos on that model reproduces T_k ; every model eigenvalue lies within the same explicit radius of $\text{spec}(H)$; and the starting spectral measures obey the dual Wasserstein and cluster-mass bounds stated above. The formal scope begins at the run interface, rather than at an executable floating-point algorithm, and that boundary should be preserved in any comparison with an informal draft.